Part 2: Pandemic

Jonathan Pincince & Shina Adewumi

1. What is the maximum number of initially infected students such that, regardless of how they are placed in the classroom, at least one initially healthy student always remains healthy?

The worst-case scenario for n students is that they are arranged in a perfect diagonal, in which case the students infect n^2 students in an n x n square. Because of the potential for this worst case, 5 students could infect 25 maximum, so you can't have 5 infected to start and have some students remain safe in all arrangements, but 4 could only infect 16 total maximum, so 4 infected students could be placed anywhere and still fail to infect the whole room.

2. What is the minimum number of initially infected students such that there is some arrangement of that many initially infected students that will result in every student eventually becoming infected?

Again by looking at the worst-case scenario, n students could lead to n x n total student infections, meaning that for 25 students, sqrt(25) = 5 students could be arranged to infect the whole room.

3. Can you arrange this minimum number of infected students in such a way that the infection never spreads to any healthy student?

This is possible in the case of all five students being in one row or column. This would mean that no other student could sit next to 2 infected students, since they would have to share a row and column, but we've already filled a row or column entirely with infected students.

4. How would your answers change if there were n^2 students in an n x n grid?

I don't think our answers would change much, since we approached this problem from a general standpoint to begin with, and the geometric properties of the diagonal leading to n^2 infections don't change when n does, as long as the classroom remains a grid. The maximum number of initially infected students is $n - 1$, the minimum number to successfully infect the whole room for some arrangement is n, and this minimum can always be arranged in such a way that the infection never spreads to any healthy student, by putting all infected students in one row or column.

5. Does it matter if n is even or odd?

Whether n is even or odd should not matter because of the nature of the worst-case, which is always a diagonal of n infected students to start. The general shape of the diagonal doesn't depend on n being even or odd, so n can be either.

6. What geometric property about the set of infected students never changes as the days pass and the infection spreads?

As long as the total number of 'edges' of the collective shape of infected students is more than 4, (the shape is not a rectangle) the infection will continue to spread. Once the total number of 'edges' of the shape is 4 (the shape is now a rectangle) the spread will stop because then no uninfected square could share an edge with two infected squares. The minimum number of edges is 4 (1 infected student).